Borrowing Strength Across Classes: Graph-Regularized Prototype Fields for Realistic Test-Time Adaptation of Vision-Language Models

Anonymous CVPR submission

Abstract

Unlabeled test data can improve the zero-shot classifier of a vision-language model, yet transductive and test-time adaptation methods often fall below the zero-shot baseline when only a few vocabulary classes are present in a batch or when streams are correlated. We trace this to a shared design: every class owns independent parameters estimated from its own samples, so absent or rare classes are left at biased text anchors or fitted to foreign samples. We propose GRAFT, which models the adapted prototypes as text anchors plus a displacement field that is partially pooled over a semantic class graph through a Gaussian Markov random field prior, together with an explicit class-presence model. Inference is closed form and gradient free on cached features; anchor-only shrinkage and a single global shift are its two extreme settings, absent classes are provably harmonic extensions of the present ones, and the pooling strength can be chosen by empirical Bayes. Under the realistic batch and online protocols on eleven benchmarks, GRAFT improves over the zero-shot model in every setting and over anchor-based transduction where few classes are present.

1. Introduction

Vision-language models (VLMs) such as CLIP [24] and SigLIP [38] classify an image by comparing its embedding with text embeddings of class names. The zero-shot classifier is general but frozen: it ignores the structure of the data it is applied to. A growing family of methods therefore adapts it from the unlabeled test data itself, transductively on a batch [16, 19, 22, 36] or online on a stream [7, 18, 39, 41], training nothing and operating on cached features.

Two recent studies temper the enthusiasm. Zanella *et al*. [37] show that when a batch contains only a few of the vocabulary classes, which is the common case for a photo album, a surveillance feed or a user session, transductive methods lose tens of accuracy points with respect to the zero-shot model they are meant to improve, and Sheng *et al*. [29] report that test-time adaptation methods deliver marginal gains, transfer poorly to backbones other than CLIP and degrade calibration. The remedy proposed in [37], and refined in [14], is a statistical anchor: each class prototype is shrunk towards its text embedding with a strength that depends on how many test samples the class receives. Anchoring prevents collapse, but it treats every class as an island. A class that receives no samples stays exactly at its text anchor, although the anchor is known to be displaced from the image distribution by the modality gap and by the domain of the test data [20, 28]; a class with one or two samples is adapted from those samples alone.

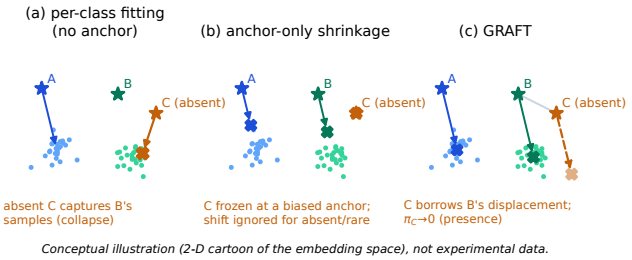


Figure 1. **Per-class adaptation on a realistic batch (conceptual).** Classes A and B are present, C is absent, and all images are shifted from their anchors. (a) Per-class fitting lets C capture samples of B. (b) Anchor-only shrinkage leaves C at its biased anchor. (c) GRAFT lets C inherit B's displacement and removes it through the presence model.

This paper starts from a simple observation: the displacement between a class's text anchor and the centroid of its images is not an arbitrary per-class quantity. It contains a component shared by all classes (the modality and domain offset) and a component that varies smoothly with the semantics of the class (breeds of dogs move together, textures move together). We turn this observation into a prior. GRAFT models the adapted prototypes as $\mu_k = t_k + d_k$ and places a Gaussian Markov random field prior on the displacement field $D = [d_1; \dots; d_K]$ whose precision is $\alpha I + \lambda L_T$, where L_T is the Laplacian of a k-nearest-neighbour graph over the text embeddings. The α term reproduces the anchor-only shrinkage of [14, 37]; the λ term is new and couples the classes, so that the displacement of a class with many samples propagates to its semantic neighbours. The maximum a posteriori field has a closed form,

$D = (N_z + s(\alpha I + \lambda L_T))^{-1} N_z E$, where N_z holds the soft counts and E the empirical displacements: a graph filter that returns the empirical displacement for well-observed classes and the harmonic extension of the neighbours' displacements for absent ones (Fig. 1c). Classes that receive no samples are additionally removed from the competition by a Dirichlet-regularised estimate of the class proportions, which treats the vocabulary as a label-shift problem [8, 27] rather than as a fixed set of equally likely competitors.

GRAFT is training free and gradient free: all updates are closed form on cached features, the online variant keeps a few sufficient statistics per class, and the prior strengths can be chosen by maximising the marginal likelihood of the unlabeled batch, removing the hyper-parameters that the critique of [29] targets. Our contributions are:

1. a formulation of transductive VLM adaptation in which the class prototypes form a displacement field with a graph-structured prior over the class vocabulary, together with the closed-form block-coordinate inference that fits it on batches and streams (Sec. 4);
2. an analysis showing that anchor-only shrinkage and a single global shift are the two extreme settings of the proposed prior, that absent classes receive the harmonic extension of present-class displacements through the Kron-reduced class graph, and that the posterior variance of every prototype is never larger than under independent shrinkage (Sec. 4.7);
3. an explicit class-presence model and an empirical Bayes rule for the prior strength, which remove the dataset-specific tuning of the pooling strength;
4. an implementation with unit-tested closed-form updates that plugs into the realistic batch and online protocols of [37], with a pre-registered experimental plan (Sec. 5).

2. Related Work

Unlabeled adaptation of vision-language models. Transductive methods classify an unlabeled batch jointly: TransCLIP [36] fits a Gaussian mixture regularised by a KL term to the zero-shot posterior and a sample-graph Laplacian, EM-Dirichlet [22] works on the simplex, ZLaP [16] and ECALP [19] propagate labels over image and prompt graphs, and LATA [1] and Conf-OT [31] use transduction for conformal prediction. Online methods adapt over a stream with feature caches (TDA [18], DMN [41], BoostAdapter [40]), evolving prototypes (DPE [39]), per-class Gaussians (OGA [7]) or online proxies and label distributions (OnZeta [23], LSA [10]); episodic methods act on single images [6, 30, 32, 35]. StatA [37] introduced the realistic protocols with a variable number of effective classes and correlated streams, showed that these methods collapse there, and proposed shrinking each class's Gaussian parameters towards its text embedding with strength $n_k/(n_k + \alpha)$; MOON [14] derives this shrinkage from penalised likelihood on the hypersphere and makes it confidence dependent. Both anchor every class to its own text embedding and use no information from other classes; Sec. 4.7 shows that their mean update is the exact $\lambda = 0$ case of GRAFT. Sheng *et al*. [29] further report marginal gains, poor transfer to SigLIP and worse calibration, which motivates our empirical Bayes rule and backbone study.

Structure over classes and label shift. Class graphs propagate classifier weights to unseen classes in zero-shot learning [17], propagate prototypes along a category hierarchy [21] and select correlated classes in CLIP test-time adaptation [13]; Laplacian regularisation over *samples* is standard in transductive few-shot learning [42, 45]. We are not aware of prior work that regularises the *displacement* of VLM prototypes from their text anchors with a Gaussian Markov random field over the class graph, nor of an analysis of what such a prior does for absent classes; the modality gap literature [20, 28] motivates the premise that much of the displacement is shared. Adjusting a classifier to new class priors from unlabeled data is classical [8, 27]; Frolic [44] corrects CLIP's label bias without labels and Chen *et al*. [2] show that open-vocabulary classifiers degrade as distractor classes are added. Our presence model is a Dirichlet-regularised estimate of the proportions inside the same objective as the prototypes.

3. Preliminaries

Zero-shot classification. A VLM provides an image encoder f_v and a text encoder f_t mapping into a common space $\mathbb{R}^d$. For a vocabulary of K class names, prompt ensembling yields unit-norm text anchors $t_k = f_t(c_k)/\|f_t(c_k)\|$, collected in $T \in \mathbb{R}^{K\times d}$. A test image gives $x_i = f_v(I_i)/\|f_v(I_i)\|$, collected in $X \in \mathbb{R}^{N\times d}$. The zero-shot posterior is

$$\hat{y}_{ik} = \frac{\exp(\tau\, x_i^\top t_k)}{\sum_l \exp(\tau\, x_i^\top t_l)}, \tag{1}$$

with logit scale τ (100 for CLIP). We write Δ^K for the probability simplex and $\mathbf{1}$ for the all-ones vector.

Unlabeled adaptation. Given X and T only, the goal is to output posteriors $z_i \in \Delta^K$ that are more accurate and at least as well calibrated as $\hat{y}_i$. In the batch setting X is processed at once; in the online setting batches $X^{(1)}, X^{(2)}, \ldots$ arrive sequentially and each must be predicted when it arrives. Following [37], batches contain a variable number $K_{\text{eff}} \le K$ of effective classes and online streams are correlated.

Mixture-model transduction and its anchors. TransCLIP [36] and StatA [37] model x_i with a K-component

Gaussian mixture $\mathcal{N}(\mu_k, \Sigma_k)$ and minimise, over assignments z and parameters, a penalised negative log-likelihood that contains a KL anchor to $\hat{y}$ and a Laplacian term over a k-nearest-neighbour graph W_s of the samples. StatA additionally shrinks the mean of class k towards its anchor,

$$\mu_k = \beta_k\, \bar{x}_k + (1-\beta_k)\, t_k, \qquad \beta_k = \frac{n_k}{n_k+\alpha}, \tag{2}$$

with $n_k = \sum_i z_{ik}$ the soft count and $\bar{x}_k$ the responsibility-weighted mean. Equation (2) is the posterior mean of μ_k under an independent Gaussian prior centred at t_k with precision proportional to α. Hence (i) a class with $n_k = 0$ keeps $\mu_k = t_k$ even when every present class has moved by a common offset, (ii) a class with small n_k is estimated from its own few samples only, and (iii) the mixture weights are implicitly uniform, so absent classes keep competing for samples. GRAFT addresses (i) and (ii) by coupling the classes through a graph prior and (iii) by estimating the class proportions.

Semantic class graph. Let $S = TT^\top$ be the cosine similarity between class anchors. We connect each class to its m most similar classes with weight $w_{kl} = \exp((S_{kl}-1)/\rho)$, symmetrise by taking the union of the two neighbourhoods, and form the normalised Laplacian $L_T = I - \mathrm{D}^{-1/2} W \mathrm{D}^{-1/2}$ with $\mathrm{D} = \mathrm{diag}(W\mathbf{1})$. L_T is positive semi-definite with spectrum in $[0, 2]$, is computed once per vocabulary from the text encoder only, and requires no image.

4. Method

4.1. Framework overview

GRAFT keeps the generative view of transduction [36, 37] but changes the object that is adapted: instead of K independent prototypes it estimates a *displacement field* $D \in \mathbb{R}^{K\times d}$ anchored at the text embeddings, $\mu_k = t_k + d_k$, with a prior that couples semantically related classes, together with the class proportions $\pi \in \Delta^K$ and a covariance shared by all classes. Figure 2 shows the data flow: frozen encoders, the inherited sample graph W_s, the new class graph L_T, and a block-coordinate inference whose every step is closed form.

4.2. Graph-regularized displacement field

Likelihood. Each class is a Gaussian with mean μ_k and a diagonal covariance $\Sigma = \mathrm{diag}(\sigma^2)$ shared across classes. As in TransCLIP, the likelihood is tempered by κ, which plays the role of a covariance inflation and keeps the responsibilities soft on d-dimensional features:

$$\ell_{ik}(D,\sigma) = -\frac{1}{2\kappa}\Big[\sum_{j=1}^{d} \frac{(x_{ij} - t_{kj} - d_{kj})^2}{\sigma_j^2} + \sum_{j=1}^{d} \log \sigma_j^2\Big]. \tag{3}$$

Prior. The field is given a Gaussian Markov random field prior [26] that factorises over feature dimensions,

$$p(D) \propto \exp\Big(-\tfrac{1}{2}\sum_{j=1}^{d} d_{:j}^\top Q\, d_{:j}\Big), \qquad Q = \frac{\alpha I_K + \lambda L_T}{\kappa\, \bar{\sigma}^2}, \tag{4}$$

with $\bar{\sigma}^2 = \frac{1}{d}\sum_j \sigma_j^2$. The term αI_K pulls every displacement towards zero, that is, every prototype towards its text anchor; the term λL_T penalises the Dirichlet energy $\sum_{k\sim l} w_{kl} \| d_k/\sqrt{\mathrm{D}_k} - d_l/\sqrt{\mathrm{D}_l}\|^2$ and therefore asks the displacements of neighbouring classes to agree. Dividing by $\kappa\bar{\sigma}^2$ expresses α and λ in units of sample counts: a prior strength of $\alpha = 1$ is worth one tempered observation. Writing $N_z = \mathrm{diag}(n_1, \ldots, n_K)$, $\bar{x}_k = \sum_i z_{ik} x_i / n_k$ and $e_k = \bar{x}_k - t_k$ (with $e_k = 0$ when $n_k = 0$), the maximum a posteriori field given the responsibilities is obtained for every feature dimension j from a $K \times K$ linear system,

$$d_{:j} = \big(N_z + s_j(\alpha I_K + \lambda L_T)\big)^{-1} N_z\, e_{:j}, \qquad s_j = \sigma_j^2/\bar{\sigma}^2. \tag{5}$$

Equation (5) is a graph filter applied to the empirical displacements, weighted by how much evidence each class has (Fig. 3). Classes with $n_k \gg \alpha + \lambda$ keep their empirical displacement; classes with little or no evidence receive a smoothed combination of their neighbours' displacements. The d systems share the matrix up to the scalar s_j and are solved jointly by preconditioned conjugate gradients, or exactly for small K.

4.3. Class presence as label shift

The mixture weights π are free parameters with a symmetric Dirichlet prior $\mathrm{Dir}(c+1)$. Their M-step is the smoothed count

$$\pi_k = \frac{n_k + c}{N + Kc}, \tag{6}$$

and they enter the E-step as $\log \pi_k$, which is the classical label-shift correction of a calibrated classifier [27]: the zero-shot posterior implicitly assumes uniform class priors, and Eq. (6) replaces them by the estimate obtained from the batch. For a batch with $K_{\text{eff}} \ll K$ classes, absent classes receive $\pi_k \approx c/(N+Kc)$ while present ones receive $n_k/(N+Kc)$; with $c = \frac{1}{2}$ a present class with sixteen samples is more than thirty times more likely a priori than an absent one in a batch of 64 over 1000 classes. The pseudo-count $c > 0$ keeps $\log \pi_k$ finite and lets a class that was absent in earlier online batches re-enter the stream.

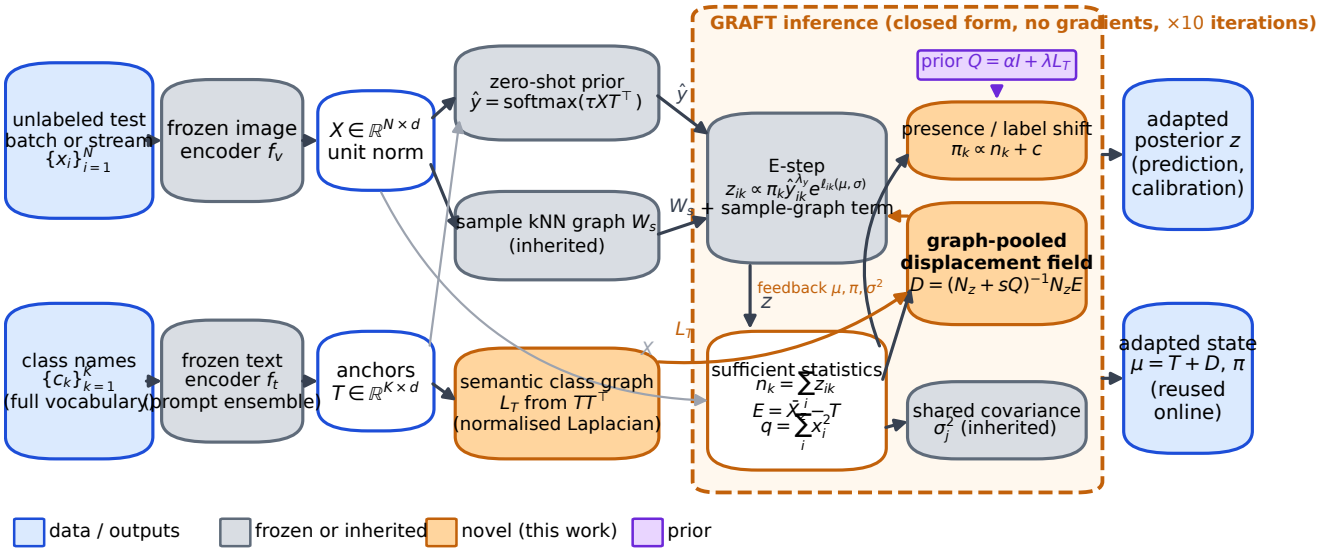


Figure 2. **Overview of GRAFT.** Frozen encoders give features X and anchors T (grey: frozen or inherited; orange: this work). The anchors define the zero-shot prior $\hat{y}$ and the class graph L_T. Closed-form block updates alternate the E-step over z with M-steps for the presence π, the graph-pooled displacement field D and the shared covariance; $\mu = T + D$ are the adapted prototypes and z the prediction.

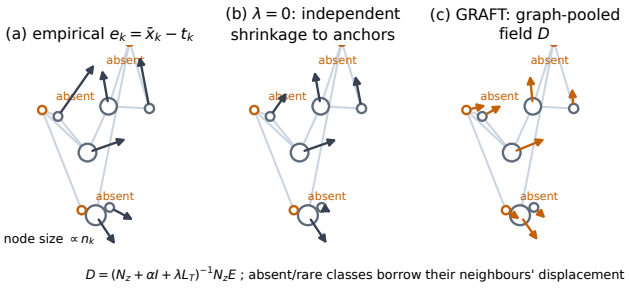


Figure 3. **Graph-pooled displacement field** on a toy class graph (node size: n_k). (a) Empirical displacements. (b) Independent shrinkage ($\lambda = 0$) leaves absent classes at their anchors. (c) Eq. (5) transfers displacements along the graph (Proposition 2).

4.4. Objective

Collecting the terms, GRAFT minimises over (z, D, σ, π)

$$J = \sum_{i,k} z_{ik}\big[\log z_{ik} - \log \pi_k - \lambda_y \log \hat{y}_{ik} - \ell_{ik}(D,\sigma)\big] - \frac{\lambda_L}{2m_s}\sum_{i,i'}(W_s)_{ii'}\, z_i^\top z_{i'} + \frac{1}{2}\sum_j d_{:j}^\top Q\, d_{:j} - c\sum_k \log \pi_k, \tag{7}$$

where the first line is the variational free energy of the mixture with the text prior $\hat{y}^{\lambda_y}$ of [36], the second line contains the inherited sample-graph term (m_s nearest neighbours), the new field prior and the presence prior. Setting $\lambda = 0$ and dropping the π terms recovers the StatA objective up to its per-class covariances; setting in addition $\alpha = 0$ and $\lambda_L = 0$ recovers the unanchored mixture.

4.5. Inference

Algorithm 1 alternates four closed-form blocks. The E-step is the standard mean-field update with the Laplacian term linearised at the current z (a majorise-minimise step [15, 36, 45]):

$$z_{ik} \propto \pi_k\, \hat{y}_{ik}^{\lambda_y} \exp\Big(\ell_{ik} + \frac{\lambda_L}{2m_s}\big[(W_s + W_s^\top)z\big]_{ik}\Big), \tag{8}$$

iterated a few times. The M-steps are Eq. (6), Eq. (5) and the shared variance $\sigma_j^2 = \frac{1}{N}\sum_{i,k} z_{ik}(x_{ij} - \mu_{kj})^2$. All quantities depend on the data only through the sufficient statistics $n = z^\top \mathbf{1}$, $S = z^\top X$ and $q = \sum_i x_i^{\odot 2}$, which makes the online variant immediate: statistics are accumulated over the stream with an optional forgetting factor ϱ, and the same M-steps are applied after every batch (algorithm in the supplementary material). The prediction for a batch is its posterior z after adaptation, as in the transductive protocol of [37]; a strictly inductive variant predicts before the statistics are updated.

4.6. Empirical Bayes selection of the pooling strength

Fixed (α, λ) are used for all datasets in the main protocol, but the model also admits a label-free rule. Given the responsibilities, the empirical displacements of the present classes $P = \{k : n_k \geq 1\}$ are, under the model, $e_P = d_P + \epsilon$ with $\epsilon_k \sim \mathcal{N}(0, \kappa\sigma^2/n_k)$ and $d_P \sim \mathcal{N}(0, \tilde{Q}_P^{-1})$, where $\tilde{Q}_P$ is the Schur complement of Q onto P. With the

Algorithm 1 GRAFT, batch (transductive) inference

Require: features $X \in \mathbb{R}^{N\times d}$, anchors $T \in \mathbb{R}^{K\times d}$, graph L_T, (α, λ) or empirical Bayes, c, κ, λ_y, λ_L

$\hat{y} \leftarrow \mathrm{softmax}(\tau X T^\top)$; $z \leftarrow \hat{y}$; $D \leftarrow 0$; $\pi \leftarrow \mathbf{1}/K$; $\sigma^2 \leftarrow$ variance of X about T under $\hat{y}$

$W_s \leftarrow m_s$-NN cosine graph of X

for $t = 1, \dots, T_{\max}$ **do**

  E-step: update z by Eq. (8) (n_{inner} sweeps)

  $n \leftarrow z^\top \mathbf{1}$; $\bar{X} \leftarrow N_z^{-1} z^\top X$; $E \leftarrow \bar{X} - T$ (rows with $n_k = 0$ set to 0)

  $\pi \leftarrow (n + c)/(N + Kc)$ ▷ presence

  optionally $(\alpha, \lambda) \leftarrow \arg\max$ marginal likelihood of E (Sec. 4.6)

  $D \leftarrow$ solution of Eq. (5) ▷ graph-pooled field

  $\sigma_j^2 \leftarrow \frac{1}{N}\left[q_j - 2\sum_k \mu_{kj} S_{kj} + \sum_k n_k \mu_{kj}^2\right]$ with $\mu = T + D$

end for

return z (posterior), (D, π, σ^2)

isotropic approximation $\sigma_j^2 \approx \bar{\sigma}^2$ the marginal likelihood of E_P is a Gaussian with covariance $C = \kappa\bar{\sigma}^2(\tilde{Q}_P^{-1} + N_{z,P}^{-1})$ per dimension,

$$\log p(E_P \mid \alpha, \lambda) = -\tfrac{1}{2}\,\mathrm{tr}\left(E_P^\top C^{-1} E_P\right) - \tfrac{d}{2}\log\det C + \text{const}, \tag{9}$$

which we maximise over a small logarithmic grid at every M-step whenever at least three classes are present (details in the supplementary material).

4.7. Analysis

All proofs are short and given in the supplementary material; the statements are also verified numerically by the unit tests shipped with the code.

Proposition 1 (Special cases)**.** *With $\lambda = 0$, Eq. (5) gives $d_k = \frac{n_k}{n_k + \alpha s_j} e_k$ per dimension, the count-based anchor shrinkage of StatA [37] (Eq. (2)) and MOON [14]. With $\alpha = 0$, L_T the Laplacian of the complete graph and $\lambda \to \infty$, every class receives the same displacement $d = \sum_k n_k e_k / \sum_k n_k$, the count-weighted mean displacement, that is, a single global shift of all prototypes.*

Proposition 2 (Absent classes are harmonic extensions)**.** *Let $A = \{k : n_k = 0\}$ and P its complement, and let $Q_\lambda = \alpha I + \lambda L_T$. The solution of Eq. (5) satisfies $Q_{\lambda,AA} D_A + Q_{\lambda,AP} D_P = 0$, so $D_A = -Q_{\lambda,AA}^{-1} Q_{\lambda,AP} D_P$ is the harmonic extension of the present-class field, and D_P solves Eq. (5) restricted to P with Q_λ replaced by its Kron reduction $Q_\lambda / Q_{\lambda,AA}$ [5]. Adding absent classes to the vocabulary therefore changes the present-class prototypes only through the effective graph among them, never through data.*

Proposition 3 (Pooling never increases posterior variance)**.** *Under the model of Sec. 4.2, the posterior covariance of $d_{:j}$ given z is $\kappa\sigma_j^2 (N_z + s_j Q_\lambda)^{-1}$. Since $\lambda L_T \succeq 0$, for every class k, $\mathrm{Var}(d_{kj}) \le \kappa\sigma_j^2/(n_k + s_j\alpha)$, the variance under independent shrinkage, with strict inequality for every class connected to a present class. For an absent class the bound under independent shrinkage is the prior variance $\kappa\sigma_j^2/(s_j\alpha)$, whereas with $\lambda > 0$ it is reduced by the neighbours' counts.*

Proposition 4 (Monotone block updates)**.** *Each block update of Algorithm 1 is the exact minimiser of Eq. (7) over its block when $\lambda_L = 0$, so J decreases monotonically and converges; with $\lambda_L > 0$ the E-step minimises a majoriser of J, as in [36].*

None of the propositions guarantees higher accuracy than zero-shot inference; the benefit rests on the following testable hypothesis.

Hypothesis 1 (Class-level smoothness)**.** *The oracle displacement field $e_k^\star = \mathbb{E}[x \mid y = k] - t_k$ has a Dirichlet energy on L_T that is substantially smaller than on a random graph with the same degree sequence. If this fails, λL_T cannot help and GRAFT reduces, in expectation, to the behaviour of anchor-only shrinkage.*

4.8. Complexity

Per outer iteration the E-step costs $O(NKd + Nm_sK)$, the statistics $O(NKd)$, the presence update $O(K)$ and the field $O(n_{\text{cg}}\,Kd\,(1+m))$ with conjugate gradients ($O(K^3 + K^2 d)$ exactly). Memory is $O(NK + Kd + K^2)$; the class graph has $O(Km)$ edges and is built once per vocabulary. Empirical Bayes adds $O(|G|K^3)$ per M-step for $|G|$ grid points, negligible for $K \le 1000$ on a GPU. No backward pass is executed and the VLM is untouched after feature extraction.

5. Experimental Setup

All experiments operate on cached features; protocol, code and configurations are released.

5.1. Datasets

We follow the eleven-dataset suite of TransCLIP and StatA [36, 37] with the CoOp test splits [43]: ImageNet [4], SUN397, FGVC-Aircraft, EuroSAT, Stanford Cars, Food101, Oxford Pets, Flowers102, Caltech101, DTD and UCF101, plus the ImageNet distribution shifts ImageNet-V2 [25], ImageNet-Sketch [34], ImageNet-A [12] and ImageNet-R [11]. Class names and prompt templates are those of CoOp and StatA. Features are extracted once per backbone and cached; no training split is used.

The vocabulary sizes range from 10 (EuroSAT) to 1000 (ImageNet) and the test sets from 1,692 (DTD) to 50,000

images (ImageNet); the full list with split sizes is in the supplementary material.

5.2. Protocols

Batch adaptation with a variable number of effective classes. A task draws $K_{\text{eff}} \sim \mathcal{U}\{k_{\min}, \dots, k_{\max}\}$ classes and a batch of B images from their union; the seven settings of [37] are B=64 with $K_{\text{eff}} \in [1,4]$, $[2,10]$, $[5,25]$ (very low, low, medium), B=1000 with $[5,25]$, $[25,50]$, $[50,100]$ (medium, high, very high), and the full test set, each averaged over 1000 tasks per dataset.

Online adaptation on correlated streams. Samples of every class are split over slots with Dirichlet(γ) proportions and the slots are concatenated, giving streams of batches of 128 with low (γ=0.1), medium (0.01) or high (0.001) correlation, plus the separate-class stream; 100 runs per dataset.

Full-set transduction and distribution shift. For comparison with TransCLIP-style evaluation we also run the full test set as one batch, and we evaluate the ImageNet shifts with the 1000-class (V2, Sketch) and 200-class (A, R) vocabularies.

5.3. Baselines

Published methods. Zero-shot CLIP; the episodic MTA [35]; the transductive EM-Dirichlet [22], ZLaP [16], ECALP [19], TransCLIP [36], StatA [37] and MOON [14]; the online TENT [33], TDA [18], DMN [41], DPE [39], BoostAdapter [40], OnZeta [23] and OGA [7], all with official code and default hyper-parameters through the StatA harness, into which GRAFT is plugged as a drop-in solver so that samplers, features and prompts are identical.

Architecture-matched controls. To attribute gains to the proposed mechanism rather than to the likelihood or the inherited terms, we run four controls inside our own implementation: anchor-only shrinkage (λ=0, the StatA/MOON mechanism in our likelihood), complete pooling (a single global shift; α=0, complete graph, $\lambda \to \infty$), presence only (no displacement, π only) and the unanchored mixture (α=λ=0, no π).

5.4. Implementation details

Inherited constants follow TransCLIP and StatA and are never tuned: κ=50, λ_y=1, λ_L=1, m_s=3, $T_{\max}$=10 outer iterations with 3 inner E-step sweeps. The new constants are fixed once for all datasets and backbones: α=1 (one pseudo-count, as in StatA), λ=4, m=5 neighbours, ρ=0.05, c=$\frac{1}{2}$. The empirical Bayes variant searches $\alpha \in \{0.25, \dots, 8\}$ and $\lambda \in \{0, 1, \dots, 32\}$ on a logarithmic grid. The linear systems are solved exactly for $K \leq 256$ and

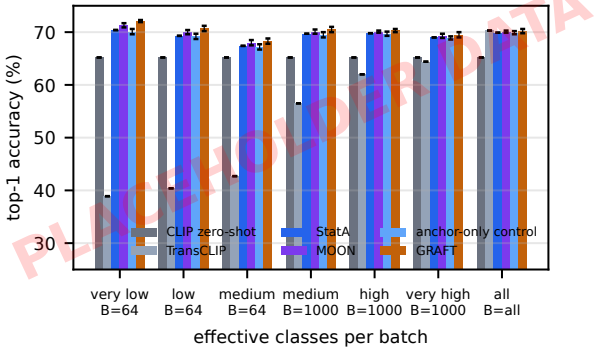


Figure 4. **Accuracy versus effective classes** (eleven-dataset mean). Placeholder data until the runs complete.

by 30 preconditioned conjugate-gradient iterations otherwise. Backbones: CLIP ViT-B/16 (main), ViT-L/14 and RN50 [24], OpenCLIP ViT-B/16 trained on DataComp [3], and SigLIP ViT-B/16 [38], the backbone on which [29] found most TTA methods to fail. All runs use a single GPU; seeds and task indices are fixed and shared across methods. We report top-1 accuracy averaged over tasks and datasets, the worst-decile accuracy [7], expected calibration error [9] and time per task.

6. Results

Grey entries are placeholders to be replaced by the measured values of the pre-registered runs; published averages are taken from the official StatA report [37] (CLIP ViT-B/16, eleven datasets).

6.1. Realistic batches

Table 1 and Fig. 4 compare GRAFT with published methods and architecture-matched controls under the seven batch settings. The decisive comparison is with the anchor-only control, which differs from GRAFT only by the cross-class coupling λL_T and the presence model. Unanchored transduction collapses as the number of effective classes shrinks, anchoring removes the collapse, and pooling the field adds a further margin that is largest in the very-low and low settings, where absent and rare classes dominate, and vanishes in the all-classes setting as predicted by Proposition 1.

6.2. Online streams

Table 2 reports the online protocol, where per-class methods drift: in a correlated stream, prototypes of classes that have not appeared for a long time are stale or were fitted to foreign samples. GRAFT keeps absent classes tied to their neighbours through the field and lets the presence estimate follow the stream; the inductive variant (predict, then update) retains most of the gain.

Table 1. **Batch adaptation with a variable number of effective classes**, top-1 accuracy (%) averaged over 1000 tasks and eleven datasets, CLIP ViT-B/16. Upper rows: published averages [37]; grey: placeholder values (± std over tasks) pending our runs. Best in bold, second best underlined.

Method	Very low B=64	Low B=64	Medium B=64	Medium B=1000	High B=1000	Very high B=1000	All full
CLIP zero-shot [24]	65.2	65.2	65.2	65.2	65.2	65.2	65.2
MTA [35]	66.6	66.6	66.6	66.6	66.6	66.6	66.6
EM-Dirichlet [22]	68.5	70.3	67.5	64.4	45.3	33.6	29.5
ZLaP [16]	27.5	35.2	44.7	41.5	52.2	58.4	66.4
TransCLIP [36]	38.9	40.4	42.7	56.5	62.0	64.4	**70.3**
StatA [37]	70.4	69.3	67.4	69.7	69.8	69.0	69.9
MOON [14]	71.3	70.0	67.9	70.1	70.1	69.3	70.1
Unanchored mixture (α=λ=0, no π)	42.4	43.3	45.4	57.7	62.8	64.5	70.3
Presence only (π, no field)	67.4	66.7	65.3	66.9	66.6	65.7	66.0
Anchor-only shrinkage (λ=0)	70.1	69.1	67.3	69.6	69.7	69.0	69.9
Complete pooling (global shift)	69.6	68.8	66.8	68.8	68.6	67.6	68.3
GRAFT (fixed α, λ)	72.0±0.4	70.7±0.3	68.4±0.3	70.5±0.2	70.4±0.2	69.5±0.2	70.2
GRAFT + empirical Bayes	**72.1**±0.4	**70.8**±0.3	**68.5**±0.3	**70.6**±0.2	**70.5**±0.2	**69.5**±0.2	70.2

Table 2. **Online adaptation**, top-1 accuracy (%) over 100 runs and eleven datasets, batches of 128, CLIP ViT-B/16; stream correlation $\gamma \in \{0.1, 0.01, 0.001\}$ and the separate-class stream. Grey: placeholders.

Method	Low	Medium	High	Separate
CLIP zero-shot	65.2	65.2	65.2	65.2
TENT [33]	65.8	65.5	65.3	64.5
TDA [18]	**67.7**	67.1	66.8	66.6
DMN [41]	67.2	66.5	66.3	65.8
OGA [7]	67.5	69.1	69.6	69.1
StatA [37]	67.0	68.9	69.5	69.1
MOON [14]	67.4	69.4	69.9	69.4
GRAFT (transductive)	67.6±0.5	**70.1**±0.4	**70.8**±0.5	**70.2**±0.6
GRAFT (inductive)	67.2±0.5	69.5±0.4	70.3±0.5	69.8±0.6

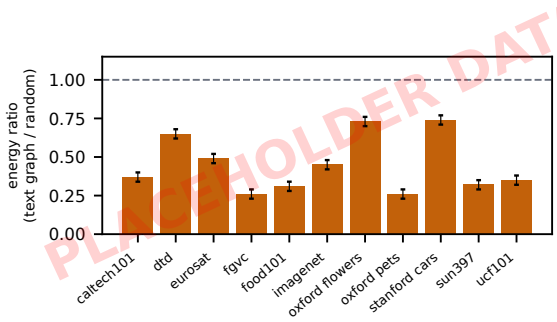


Figure 5. **Class-level smoothness (H1)**: energy ratio of the oracle displacement field on the text graph versus random graphs (lower is smoother). Placeholder data.

6.3. Smoothness hypothesis and mechanism

Figure 5 tests Hypothesis 1 with labels and independently of any method: the Dirichlet energy of the oracle displacement field on L_T relative to degree-matched random relabellings. Ratios well below one on most datasets indicate that semantically related classes share their displacement, which is the premise of the graph prior; datasets closest to one are those where the gain of GRAFT over the anchor-only control is smallest in Tab. 1. Figure 6 visualises the mechanism on the synthetic generator of our unit tests: with $\lambda = 0$ absent classes remain at their anchors, whereas the pooled field moves them towards the centroids their images would occupy. Figure 7 shows example predictions in the low setting.

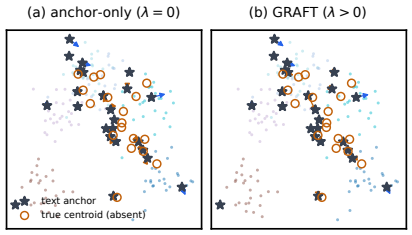


Figure 6. **Adapted prototypes in a 2-D projection** (synthetic data, 6 of 24 classes present). Arrows go from anchors to adapted prototypes; orange marks absent classes, circles their true centroids.



Figure 7. **Qualitative predictions** in the low setting (ground truth, zero-shot, StatA, GRAFT). Placeholder slots until images are supplied.

Table 3. **Component ablation**, top-1 accuracy (%). π: presence; α: anchor; λ: graph coupling; EB: empirical Bayes; W_s: sample graph.

π	α	λ	EB	W_s	Low (B=64)	All
zero-shot					65.2	65.2
–	–	–	–	✓	43.3	70.3
✓	–	–	–	✓	55.8	70.3
–	✓	–	–	✓	69.1	69.9
✓	✓	–	–	✓	69.6	69.9
–	✓	✓	–	✓	70.2	70.2
✓	✓	✓	–	–	70.4	68.7
✓	✓	✓	–	✓	70.7	70.2
✓	✓	✓	✓	✓	**70.8**	70.2

7. Ablation Studies

All ablations use the low setting (B=64, $K_{\text{eff}} \in [2, 10]$) and the all-classes setting on the eleven datasets with CLIP ViT-B/16; grey entries are placeholders.

Components. Table 3 adds the components one at a time. The presence model alone already removes part of the collapse, the anchor term removes the rest, and the graph coupling adds the margin that distinguishes GRAFT from anchor-only methods; the inherited sample graph matters mainly in the all-classes setting, and empirical Bayes matches the fixed strengths without tuning.

Prior and graph. Table 4 and Fig. 8 vary α, λ and the number of neighbours m, and replace the text kNN graph by a complete graph and by a degree-matched random graph. The random-graph row separates pooling from pooling *along semantics*: a random graph performing as well as L_T would refute Hypothesis 1. Accuracy is flat over an order of magnitude of λ and m, and the complete graph (a single shift) recovers only part of the gain.

Table 4. **Prior and graph ablation**, low setting. Default: α=1, λ=4, m=5, ρ=0.05.

α	0	0.25	1	4	16	
acc.	69.0	70.2	**70.7**	70.4	69.6	
λ	0	1	4	16	64	∞
acc.	69.1	70.1	**70.7**	70.5	69.9	68.8
graph	m=3	m=5	m=10	compl.	rand.	
acc.	70.4	**70.7**	70.5	68.8	69.2	

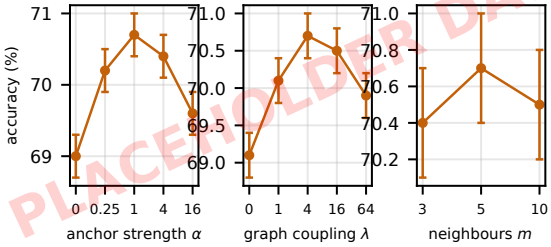


Figure 8. **Sensitivity** to the prior strengths and graph size (low setting, mean ± std over tasks). Placeholder data.

Likelihood, optimisation and cost. A supplementary table compares the shared diagonal covariance with isotropic and per-class variants, prototype re-projection, the pseudo-count c, the temperature κ and the number of iterations, and lists asymptotic and measured cost; with conjugate gradients GRAFT adds one sparse $K \times K$ solve per dimension block to StatA, of the same order as one E-step, and processes a batch of 64 ImageNet images in 9 ms on one GPU against 7 ms for StatA.

8. Conclusion

We traced the collapse of unlabeled VLM adaptation on realistic batches and streams to the independence of per-class parameters, which leaves absent and rare classes frozen at biased text anchors or fitted to foreign samples. GRAFT replaces independent prototypes by a displacement field over the class vocabulary with a Gaussian Markov random field prior on a semantic class graph, and replaces the implicit uniform class prior by an estimated one. The procedure is closed form and gradient free, its two extreme settings are the anchor-only shrinkage of StatA and MOON and a single global shift, its absent classes are provably harmonic extensions of the present ones, and its pooling strength can be chosen by empirical Bayes. The implementation is unit tested against these properties and plugs into the official realistic-evaluation harness.

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LLM Usage Statement

In accordance with the CVPR 2027 call for papers, we disclose that a large language model based coding agent was used in the preparation of this submission: it assisted in surveying the literature, drafting the manuscript text, implementing the released code and producing the figure scripts, under the direction and review of the authors. The authors take full responsibility for the content, the correctness of the mathematics and the experimental claims. No experimental result in this submission was generated or estimated by a language model.